they feel things are bound to get difficult as digital effects get more complex. But, Facebook expects that the future hardware enhancements will surely boost their machine learning models.

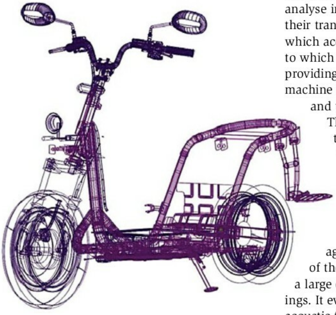
CES 2017 was full of AI-powered consumer products. Apart from the expected dominance of AI-enabled smartphones, wearable devices, home appliances and cars, two interesting developments relate to operating systems and home assistants. According to industry experts, more than 40 million homes will have a home assistant by 2021. In November last year, Google launched Google Home for this segment. Amazon's Alexa is too well-known to be introduced again. Samsung has come up with Otto, and Bosch is backing Kuri. Facebook too demonstrated a personal assistant, though there is no information about its availability yet. Apple is also supposedly working on a Siri-powered home assistant.

With so many connected and intelligent devices all around us, people are getting very worried about privacy and data safety. So, companies like Google and Norton have also come up with solutions to secure your devices. Google is offering Android Things—an operating system that powers smart devices and the Internet of Things (IoT) in a secure way. Norton Core is a mobile-enabled Wi-Fi router equipped with machine learning and Symantec's threat intelligence techniques to defend your home network from potential threats.

Lots more on the anvil

Researchers all over the world are still exploring the possibilities of AI. What was fiction a decade ago has become real now, and fiction today is being chiselled into reality at labs across the world—and by start-ups too.

So far, AI has been achieved mainly using complex algorithms. Now, imagine a chip that by itself works like a synapse of the human brain—wouldn't it make AI more real and human-like than ever before? Researchers at CNRS, Thales, have managed to create directly on a chip an artificial synapse that is capable of learning. You can use these chips to create intelligent systems comprising a network of synapses, requiring much less time and energy.



Twenty Two Motors' smart scooter for Indian roads (Courtesy: Twenty Two Motors)

Another system developed at the Sandia National Laboratories aims to improve the accuracy with which cybersecurity threats (or bad apples) are detected. The brain-inspired Neuromorphic Cyber Microscope designed by the lab can look for complex patterns that indicate specific bad apples, consuming less power than a standard 60-watt light bulb. This small processor was found to be more than a hundred times faster and a thousand times more energy-efficient than racks of conventional cybersecurity systems.

Lots of interesting AI research is happening at MIT too. Last month, MIT researchers presented a paper proposing a fast and inexpensive way to achieve speech recognition. Current speech recognition systems require a computer to analyse innumerable audio files and their transcriptions, to understand which acoustic features correspond to which typed words. However, providing these transcripts to the machine learning system is a costly and time-consuming affair.

This limits speech recognition to a small number of languages. The new approach proposed by the researchers does not rely on transcripts. Instead, the system analyses the correlation between images and spoken descriptions of those images, as captured in a large collection of audio recordings. It eventually learns which acoustic features of the recordings correlate with which image characteristics. According to the scientists, this is more natural—more like the way humans learn. Plus, it is less expensive and less time-consuming, opening up the possibility of extending speech recognition to a larger number of languages.

Another research at MIT attempts to make machines predict the future. For humans, it is easy to understand what will happen when a player bowls a ball or when a car skids off the road. However, it is not so easy for machines. A team at MIT is developing a deep-learning algorithm for this. From a still image of a scene, the system can create a short video simulating the future of that scene. Researchers feel that such generative videos can be used to add animation to still images, detect anomalies in security footage, compress data for storing and sending longer videos, and so on.